

Accounts Service Architecture

VistaPort Media Inc. — Product Data and Technology

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This document describes the architecture of VistaPort's authentication, session and account services as at the date of writing. It is an engineering reference for the Platform team and is used for onboarding and change review. It does not describe any incident.

1. Component overview

Component	Responsibility
VPAuth	Authentication and session-token service; issues and validates session cookies signed with the active signing key set.
SSOBridge	Single sign-on federation across VistaPort properties; relies on VPAuth for the authenticated-user assertion.
AtlasID	Identity graph and cross-property account linkage / match service.
legacy_uap	Legacy Unified Account Profile store (pre-2022 identity/profile data); feeds remaining linked attributes into AtlasID; scheduled for decommission.
Snowcap	Analytics warehouse holding auth and engagement event tables.

2. Authentication & session flow

- 1 A user authenticates against **VPAuth**, which verifies the credential against the account store.
- 2 VPAuth issues a **session cookie** signed with the current signing key. Each signed cookie carries a key identifier (kid) so that validators know which key to use.
- 3 Properties and **SSOBridge** validate the session cookie's signature on each request; a valid signature establishes the authenticated session.
- 4 Identity resolution for personalisation and addressability is performed by **AtlasID** using the authenticated-user key.

3. Session-cookie signing and key management

Session cookies are signed using an HMAC/asymmetric signing key referenced by a key identifier (kid). The validator trusts any cookie whose signature verifies against a key in the active key set. The active set is intended to include the current key plus, for backward compatibility during migrations, one or more legacy keys.

Key id (kid)	Role	Rotation policy	Notes
vpauth-current	Active signing key	Rotate every 12 months	Primary key for newly issued cookies
vpauth-legacy-2019	Legacy validation key (NOT used to sign new cookies)	Overdue — unrotated since 2019	Retained in the active set for backward compatibility only; pending rotation/retirement
vpauth-mobile	Mobile session key	Rotate every 12 months	Mobile clients

Rotation policy: signing keys should be rotated on a fixed cadence and any key retained only for backward compatibility should be retired promptly once the dependent migration completes. Trust in a signed cookie is only as strong as the confidentiality of the signing key referenced by its kid.

4. legacy_uap and migration

The **legacy_uap** store predates the current account platform and continues to feed selected attributes into AtlasID during migration. Decommissioning legacy_uap and retiring the associated legacy signing key (kid *vpauth-legacy-2019*) are tracked under the platform-health roadmap (DR-047) and the decommission plan (DR-057).

5. Dependencies

- SSOBridge depends on VPAuth signature validation; key-set changes must be coordinated.
- AtlasID depends on legacy_uap lineage for a subset of match keys until migration completes.
- Analytics (Snowcap) ingests auth/session events for monitoring and reconciliation.

Prepared by Nina Petrov, Head of Security Engineering. Engineering reference — confidential.